

SAMMAM MAHDI

Computer Science Educator & Researcher · Machine Learning · Algorithms

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📍 Dhaka, Bangladesh

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Summary

Computer Science graduate from BRAC University (CGPA **3.92/4.00**, Highest Distinction) with hands-on **teaching experience** and **four peer-reviewed IEEE publications**. As a Teaching Assistant I mentored **100+ students** and led weekly recitations in Data Structures & Algorithms, and I am eager to bring the same clarity to both **theory and laboratory courses** as faculty. Strong foundations across algorithms and machine learning, with a genuine passion for explaining complex ideas simply and helping students build real, working skills.

Research Interests

Trustworthy & Explainable AI · Computer Vision (3D vision, SLAM, human-computer interaction) · Deepfake Detection & Media Forensics · Assistive Technology & Accessibility · Applied Machine Learning for Healthcare · Representation & Self-Supervised Learning · Efficient, Real-Time Deep Learning.

Education

Bachelor of Science in Computer Science and Engineering

Graduated June 2026

BRAC University, Dhaka, Bangladesh

CGPA: 3.92 / 4.00 — **Honors:** Highest Distinction, Multiple Dean's List Awards

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Computer Networks, Machine Learning, Computer Vision, Software Engineering.

International Advanced Levels (IAL)

2019 – 2021

Edexcel — Average Grade A

International General Certificate of Secondary Education (IGCSE)

2018 – 2019

Edexcel — Average Grade A

Teaching Experience

Teaching Assistant — Data Structures & Algorithms

Jan 2025 – May 2026

CSE Department, BRAC University

- Mentored **100+ undergraduate students**; led weekly recitations covering algorithm complexity, graph algorithms, and dynamic programming.
- Conducted problem-solving sessions, evaluated assessments, and prepared supplementary explainer materials and worked examples.

Teaching Interests & Courses Able to Teach

Programming & Foundations: C, C++, Java, Python; Object-Oriented Programming; Data Structures & Algorithms

Systems: Operating Systems; (intro) Database Systems

AI & Data: Introduction to Artificial Intelligence; Machine Learning; Computer Vision; Deep Learning

Publications

Total citations: 29 (Google Scholar, 2026)

- **Sammam Mahdi**, RU Karim, T Sujat, SM Tabassum, A Samin, AN Zereen. “Vision-based Hand Gesture Virtual Keyboard-Mouse Framework with Bilingual Next-word Prediction.” *2026 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE)*, Bangkok, Thailand, 2026. [DOI] [1st Author].
- RU Karim, **S. Mahdi**, A Samin, AN Zereen, M Abdullah-Al-Wadud, J Uddin. “Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis.” *IEEE Access*, vol. 13, pp. 32636–32660, 2025 (IF 3.9). [DOI] — **Cited 24×** [2nd Author].
Contribution: Engineered the XAI interpretability pipeline using feature importance over 228 MediaPipe facial landmarks; co-designed the Multimodal Voting Classifier (RF + XGBoost + CatBoost), achieving 94.75% accuracy on stroke/non-stroke classification.
- T Ferdous, RU Karim, A Samin, **S. Mahdi**, AN Zereen. “Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning.” *IEEE 2nd Intl. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, 2024 [4th Author].
- T Ferdous, RU Karim, A Samin, **S. Mahdi**, H Tasnim, AN Zereen. “Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification.” *IEEE 2nd Intl. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, 2024 [4th Author].

Research Experience

DRISTEE — Agentic Assistive Vision System (Undergraduate Thesis)

2025 – 2026

Undergraduate Researcher, BRAC University

- Built a modular **agentic** perception pipeline for the visually impaired, combining **depth-based 3D object detection (CAMDet)**, **MASt3R-SLAM** dense metric mapping, and real-time scene understanding with contextual interaction.
- Integrated multimodal components — **VLM-based OCR** (Pixtral-12B) for scene-text/document reading, **face recognition** (AdaFace + RetinaFace), and a **low-latency speech interface** with **semantic caching** for responsive, context-aware assistive navigation.

ICEA — Invariant-Counterfactual Evidence Accumulation for Deepfake Detection 2025 – present

Undergraduate Researcher, BRAC University

- Designing a physics-grounded deepfake detection framework that identifies violations of five human invariant laws rather than generator-specific artifacts, enabling generalization to unseen synthetic media.
- Developing a dual-encoder backbone with a Counterfactual Violation Classifier for multi-label forensic attribution and grounded explainability.

BRACU Mongol Tori — Mars Rover, AI/ML Contributor

2023 – 2025

Two-time Participant, University Rover Challenge (URC), Utah, USA

- Designed and trained YOLOv8 models for real-time detection of bottles, mallets, and keyboard keys under outdoor lighting variability; optimized for on-device inference on NVIDIA Jetson.

Technical Skills

Deep Learning:	PyTorch, TensorFlow, Hugging Face Transformers, Mamba-2, MViTv2, BYOL-A, SHAP, GradCAM
Computer Vision:	OpenCV, MediaPipe, YOLOv5/v8, CAMDet, AdaFace, RetinaFace, Pixtral-12B, CLAHE, SLAM
Robotics / Embedded:	ROS / ROS2, Arduino, NVIDIA Jetson (Nano, Xavier NX)
Systems & Dev:	C, C++, Python (Expert), JavaScript, SQL, Linux, Git, Docker, L ^A T _E X

Selected Projects

Die Again — 3D Troll Platformer (Computer Graphics & Game Design)

[Code](#)

JavaScript, Three.js / WebGL, React — Designed and built a 3D platformer-puzzle game with real-time WebGL rendering (Three.js + react-three-fiber, post-processing), 10 themed levels, procedural Web Audio, Firebase cloud leaderboards & OAuth, and a cross-platform Tauri desktop build.

SkillWise: AI-Powered E-Learning & Anti-Cheat System

[Code](#)

MERN + Computer Vision — Designed an AI proctoring module using **MediaPipe** facial-landmark detection to flag academic dishonesty in real-time; **3rd Runner-up**, Software Engineering Showcase (200+ teams).

Metadata Journaling File System

[Code](#)

C, Operating Systems — Implemented an ext2-inspired journaling file system from scratch with crash-consistent metadata writes, inode management, and block-level journaling (useful for OS lab teaching).

LibroBot: RFID-Based Library Automation Robot

[Code](#)

Arduino, ROS, Robotics — Built an autonomous book-retrieval robot using RFID localization and DC motor control for embedded library automation.

Honors & Awards

- **Highest Distinction & Dean's List** (multiple semesters) — BRAC University; CGPA 3.92/4.00.
- **Daily Star & Edexcel High Achievers' Award** — national recognition for O/A-Level performance, Bangladesh.
- **HULT Prize Semi-Finalist** — competitive international social-entrepreneurship competition.
- **National Mathematical Olympiad Participant** — selected among top school-level competitors.
- **Duke of Edinburgh Bronze Award** — leadership and community service.

References

Dr. Aniqua Zereen	aniqua@bracu.ac.bd — Assistant Professor, CSE, BRAC University
Dr. Farig Sadeque	farig.sadeque@bracu.ac.bd — Assistant Professor, CSE, BRAC University
Dr. Md. Khalilur Rahman	khalilur@bracu.ac.bd — Associate Professor, CSE, BRAC University